

## CCNA 200-301 - Chapter 20

### OSPF (Open Shortest Path First)

#### Overview

OSPF is a Standard, Link-State, Classless dynamic routing protocol used to find the best path between networks.

#### Features

- Standard Protocol
- Classless Protocol (supports VLSM)
- Multicast: 224.0.0.5
- Administrative Distance = 110
- Metric = Cost =  $10^8 / \text{Bandwidth}$
- Loop-Free Topology
- Unlimited Hop Count

#### Comparison

| Protocol | Metric                      | AD  | Multicast  | Notes      |
|----------|-----------------------------|-----|------------|------------|
| RIP      | Hop Count                   | 120 | 224.0.0.9  | 15 hops    |
| EIGRP    | Bandwidth+Delay+Reliability | 90  | 224.0.0.10 | Hybrid     |
| OSPF     | Cost                        | 110 | 224.0.0.5  | Link-State |

#### OSPF Workflow

1. Hello Packet
2. Neighbor Discovery
3. Exchange LSA
4. Flood received LSA after saving a copy
5. Build LSDB
6. Build SPF Tree
7. Run Dijkstra (SPF)
8. Create Routing Table

#### Wildcard Mask

Wildcard = Inverse of Subnet Mask

| Subnet Mask   | Wildcard  |
|---------------|-----------|
| 255.255.255.0 | 0.0.0.255 |

255.255.0.0

0.0.255.255

255.0.0.0

0.255.255.255

### Process ID vs AS

OSPF Process ID is local and does not need to match across routers.

EIGRP Autonomous System must match across all routers.

### Important Commands

enable

configure terminal

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.255.255.255 area 0

network 11.0.0.0 0.255.255.255 area 0

network 12.0.0.0 0.255.255.255 area 0

show ip route

show ip ospf neighbor

show ip protocols

show ip ospf database

### Exam Notes

- Hello discovers neighbors.
- LSA is exchanged instead of the full routing table.
- Each router floods LSAs after saving a copy.
- Area 0 is the Backbone Area.
- OSPF uses Dijkstra SPF algorithm.